Interferometric X-Band Observations of the Sun

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We present interferometric observations of the Sun at 10.45 GHz using the two-dish correlating interferometer at the New Campbell Hall (NCH) observatory, Berkeley. Five independent baseline-determination methods—including short-time Fourier transform (STFT) fringe-frequency fitting, delay-spectrum analysis, nonlinear least-squares (NLS) fringe fitting, and a two-dimensional brute-force grid search—are applied to horizon-to-horizon drift-scan data comprising 8226 visibility integrations across 285 spectral channels. The adopted baseline, $b_{\rm ew} = 15.1148 \pm 0.0000$ m (formal) and $b_{\rm ns} = 1.4746 \pm 0.0000$ m, is consistent with an independent iPhone LiDAR survey (15.17 ± 0.30 m) at the 0.2σ level. The angular diameter of the Sun is measured from the Bessel-function (jinc) modulation of the fringe-amplitude envelope, yielding $D_\odot = 34.664 \pm 0.034'$ (statistical) $\pm 0.66'$ (systematic), consistent with the 17 GHz radio reference of $32.55 \pm 0.05'$ (C. L. Selhorst et al. 2004). A non-zero sunspot floor $f \approx 0.033$ and a limb-brightening coefficient $\varepsilon \approx 2.0$ are detected, indicating an active region on the solar disk during the observation.

1. Introduction

Radio interferometry is the foundation of modern radio astronomy and the technique by which the event horizon of a supermassive black hole was first imaged. By correlating the signals from two or more spatially separated antennas, an interferometer achieves angular resolution not of the diameter of any single dish, but by its baseline separation.

In this report, we use the two-dish X-band interferometer at the New Campbell Hall (NCH) to explore three scientific goals: **(1)** calibration of the baseline vector $(b_{\rm ew}, b_{\rm ns})$ via 5 independent methods; **(2)** measurement of the solar diameter, which we expect to contain emissions beyond the optical photosphere into the chromosphere; and **(3)** indirect detection of unresolved sunspot emissions as a non-zero visibility amplitude floor at Bessel-function nulls.

2. Background Theory

A single radio dish of diameter D has an angular resolution diffraction-limited to $\theta \sim \lambda/D$. At our observing wavelength of $\lambda = 2.87$ cm, a 1-metre dish resolves only $\sim$1.6° which would be too coarse to resolve the Sun's $\sim$0.5° disk, much less finer structures. A 15-metre dish would be needed to match the Sun's angular size, and arcsecond resolution would require a dish hundreds of metres across.

Interferometry circumvents this; instead of building one enormous aperture, we place two dishes a distance b (baseline) apart and correlate their signals. The correlated output is sensitive to angular structure on the scale $\theta \sim \lambda/b$, equivalent to a full dish of aperture diameter b. For a 15-metre baseline at 2.87 cm, this corresponds to $\theta \sim 6.6'$, in the order of the solar diameter and sufficient to resolve any disk structure.

Two dishes separated by baseline b observe the Sun at an angle θ from the zenith. Plane wavefronts from the source arrive at the two dishes with a relative path delay τ_g, which depends on the projected baseline and changes as the Earth rotates. The signals are brought together in the SNAP board, which multiplies the Fourier-transformed voltages to produce the complex visibility spectrum. Figure 1 illustrates the geometry.

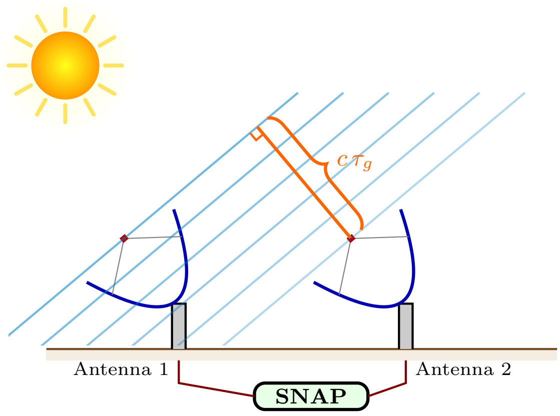

Fig. 1.— Illustration of a two-dish correlating interferometer. Two dishes separated by baseline b observe an extended source. Plane wavefronts arrive at the two antennas with a relative geometric delay. The signals are brought to the SNAP board, which digitises, Fourier-transforms, and multiplies the voltages to produce the complex visibility spectrum $V_{12}(\nu, t)$. As the Earth rotates, τ_g changes, and the correlated output traces out the interferometric fringe.

Rather than measuring brightness, an interferometer measures the complex visibility $V(u)$, which is the Fourier component of the sky brightness distribution at the spatial frequency $u = b_{\rm proj}/\lambda$, where $b_{\rm proj}$ is the projected baseline perpendicular to the line of sight.

Here, $b_{\rm proj} = \sqrt{|\mathbf{b}|^2 - (\mathbf{b}\cdot\hat{\mathbf{s}})^2}$, where $\hat{\mathbf{s}}$ is the unit vector toward the source. The component along the line of sight, $\mathbf{b}\cdot\hat{\mathbf{s}} = c\,\tau_g$, is the geometric delay.

From a fixed position, the projected baseline changes continuously as the Earth rotates, and an interferometer observing an object samples a different spatial frequency. Over a full horizon to horizon observation, u varies and a single baseline traces out an arc in the Fourier plane. By combining visibility measurements at many spatial frequencies, one can reconstruct the image of the source.

For a point source, the visibility amplitude is constant and the phase oscillates with the changing geometric delay producing fringes; for an extended source like the Sun, the visibility amplitude is modulated by the Fourier transform of its shape, producing characteristic nulls where the fringe spacing matches the source size (Section 2.4). The positions of these nulls encode the source diameter.

2.1. Path Delay

We decompose the baseline into an east–west component $b_{\rm ew}$ and a north–south component $b_{\rm ns}$. For a source at hour angle h and declination δ, observed from latitude L, the east–west baseline contributes a geometric delay

$$\tau_{g,\rm ew}(h) = \frac{b_{\rm ew}}{c}\cos\delta\,\sin h, \tag{1}$$

and the north–south baseline contributes

$$\tau_{g,\rm ns}(h) = \frac{b_{\rm ns}}{c}\sin L\,\cos\delta\,\cos h - \frac{b_{\rm ns}}{c}\cos L\,\sin\delta. \tag{2}$$

Since the second term is independent of h, we absorb this constant NS term into the instrumental cable delay τ_c, defining an effective geometric delay

$$\tau_g'(h) = \frac{b_{\rm ew}}{c}\cos\delta\,\sin h + \frac{b_{\rm ns}}{c}\sin L\,\cos\delta\,\cos h \tag{3}$$

and a modified cable delay

$$\tau_c' = \tau_c - \frac{b_{\rm ns}}{c}\cos L\,\sin\delta. \tag{4}$$

The total relative delay between the two signal paths is $\tau_{\rm tot} = \tau_g'(h) + \tau_c'$.

2.2. The Point-Source Fringe

For a monochromatic point source of frequency ν, the voltages at the two antennas are

$$E_1(t) = \cos(2\pi\nu t), \qquad E_2(t) = \cos\!\big(2\pi\nu\,[t+\tau_{\rm tot}]\big). \tag{5}$$

The aforementioned correlator output is proportional to $\langle E_1(t)\,E_2(t)\rangle$, the time-averaged product of the two signals. Together, the fringe output is

$$\begin{aligned} F(h) &= \cos\!\big(2\pi\nu\,[\tau_g'(h) + \tau_c']\big) \\ &= A\cos\!\big(2\pi\nu\,\tau_g'\big) + B\sin\!\big(2\pi\nu\,\tau_g'\big), \end{aligned} \tag{6}$$

where $A = \cos(2\pi\nu\,\tau_c')$ and $B = -\sin(2\pi\nu\,\tau_c')$ encode the instrumental phase. The unknown baseline parameters appear only inside τ_g', while A and B are linear and can be solved analytically, making this a separable nonlinear least-squares problem.

Physically, the interferometer projects a sinusoidal pattern onto the sky with fringe spacing $\lambda/b_{\rm proj}$, producing the oscillating visibility. Equivalently, the two antennas act as a giant double-sideband mixer, using the geometric delay rate to down-convert spatial structure as a temporal fringe. For extended sources, the output depends on how the source overlay the fringes: when the fringe spacing matches the source diameter, positive and negative lobes cancel and the visibility vanishes (Figure 2).

Fig. 2.— Fringe pattern projected on the sky for an extended source. *Left:* Short projected baseline produces wide fringes; the disk fits within one lobe and the visibility is high. *Right:* Long projected baseline produces narrow fringes; the disk spans multiple lobes whose + and − contributions partially cancel, reducing the visibility. At a Bessel null the cancellation is exact.

2.3. Fringe Frequency

The fringe phase $\phi(h) = 2\pi\nu\,\tau_g'(h)$ varies with hour angle. Differentiating the fringe phase with respect to time gives the instantaneous fringe frequency:

$$f_f(h) = \omega_\oplus\left[\frac{b_{\rm ew}}{\lambda}\cos\delta\,\cos h - \frac{b_{\rm ns}}{\lambda}\sin L\,\cos\delta\,\sin h\right], \tag{7}$$

where $\omega_\oplus = 7.2921\times10^{-5}$ rad s^{-1} is the sidereal rotation rate. For our baseline at $\delta\approx 0°$, the fringe period at the meridian is ∼26 s. Since $f_f(h)$ varies slowly, a short-time Fourier transform (STFT) recovers it for every h bin, and fitting the measured $f_f(h)$ curve to equation (7) yields a baseline estimate (Method 1, §4).

For an extended source with brightness distribution $I(\Delta h)$, the interferometer response is the convolution of the point-source fringe with the source profile. For a symmetric source this factors as:

$$R(h_s) = F(h_s)\times\int I(\Delta h)\,\cos(2\pi f_f\,\Delta h)\,\mathrm{d}\Delta h. \tag{8}$$

where the integral adds a slow-varying modulation to the fringe pattern.

2.4. Uniform Disk Visibility

Modelling the Sun as a uniform disk of radius R, the modulating function evaluates to the jinc:

$$\frac{V(|u|)}{V(0)} = \frac{2\,J_1(2\pi\,|u|\,R)}{2\pi\,|u|\,R}, \tag{9}$$

with zeros at $|u|\,R = j_{1,k}/(2\pi)$, where $j_{1,k}$ are the zeros of J_1. Differentiating with respect to $|u|$, the extrema of $|V|$ satisfy

$$\frac{\mathrm{d}}{\mathrm{d}|u|}\left[\frac{V}{V(0)}\right] = -\frac{2\,J_2(2\pi|u|R)}{|u|} = 0, \qquad (10)$$

so the minima (nulls) occur at the J_1 zeros $j_{1,k}$ and the maxima (sidelobe peaks) at the J_2 zeros $j_{2,k}$. Both provide independent radius estimates:

$$R = \frac{j_{n,k}}{2\pi\,|u_k|}, \qquad (11)$$

where $j_{n,k}$ is the appropriate J_1 or J_2 zero for the k-th identified extremum.

3. Instrument and Observations

3.1. The X-Band Interferometer

The NCH interferometer consists of two dishes on the roof of Campbell Hall ($L = 37.8732°$ N, 122.2571° W), with a nearly east–west baseline. Each dish feeds a double-conversion heterodyne receiver: LO1 at 8.750 GHz converts the sky signal from the RF band (centred at 10.450 GHz) to an IF near 1.7 GHz via a DSB mixer, and LO2 at 1.545 GHz down-converts the IF to baseband via an SSB mixer. A K&L Microwave bandpass filter (5B120-155/110-B/B) at the output of each baseband chain limits the usable bandwidth to ∼70 MHz (sky: 10.415–10.485 GHz), yielding 285 usable channels (ch 513–798) out of the full 250 MHz Nyquist band.

A cross-correlator implemented on a SNAP signal-processing board (CASPER 2024) digitises both baseband signals at 500 Msps, computes a 2048-point real FFT (yielding 1024 spectral channels with spacing $\Delta\nu = F_s/N_{\mathrm{FFT}} = 244$ kHz), and multiplies the resulting complex voltage spectra. The cross-correlation is integrated over 305,200 spectra, corresponding to an integration time of 1.25 s per visibility.

Figure 3 shows a summary block diagram of the signal path for one antenna; the full detailed diagram is available online[1].

An independent baseline survey was performed using the LiDAR sensor of an iPhone 13 Pro, yielding $b_{\mathrm{ew}}^{\mathrm{lidar}} = 15.17 \pm 0.30$ m and $b_{\mathrm{ns}}^{\mathrm{lidar}} = 1.36 \pm 0.30$ m. The conservative 30 cm uncertainty encompasses the difficulty of identifying the true cardinal pointing of each baseline vector.

3.2. Observations

Horizon-to-horizon drift-scan observations of the Sun were obtained on 2026 March 20 (UTC). The data span hour angles from $-79.9°$ to $+22.0°$, totalling 8226 visibility integrations divided across three observation CHIPS (Contiguous Horizon-to-horizon Integration Period Segments). The Sun's declination ranged from $\delta = -0.144°$ to $-0.032°$ over the observation. Table 1 summarises the observation log.

Table 1: Observation log.

CHIPS	Captures	HA range [deg]	Duration [hr]
0	5514	−79.9 to −8.0	4.8
1	403	−7.7 to −5.3	0.2
2	2309	−3.3 to +22.0	1.7

Channels at multiples of 256 (indices 0, 256, 512, 768) are contaminated by DC and FPGA pipeline harmonics and are excluded from all analysis. Residual ADC offsets in the SNAP board introduce a slowly

[1] https://github.com/AaronParsons/ugradio/blob/main/lab_interf/interf_signal_path.xcf

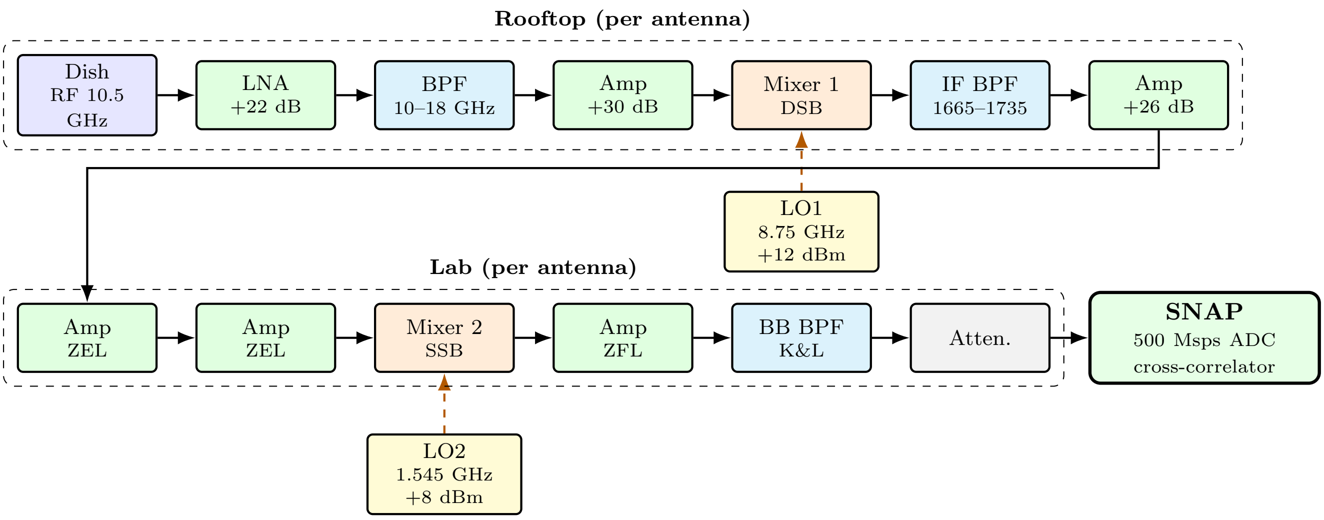

Fig. 3.— Summary signal path for one antenna of the NCH interferometer. The X-band RF signal (∼10.5 GHz) is amplified, filtered, and down-converted to IF (∼1.7 GHz) by DSB Mixer 1 with LO1 at 8.75 GHz. In the lab, SSB Mixer 2 with LO2 at 1.545 GHz converts to baseband (0–250 MHz) for digitisation by the SNAP board. Both antennas share the same LO1 and LO2 sources.

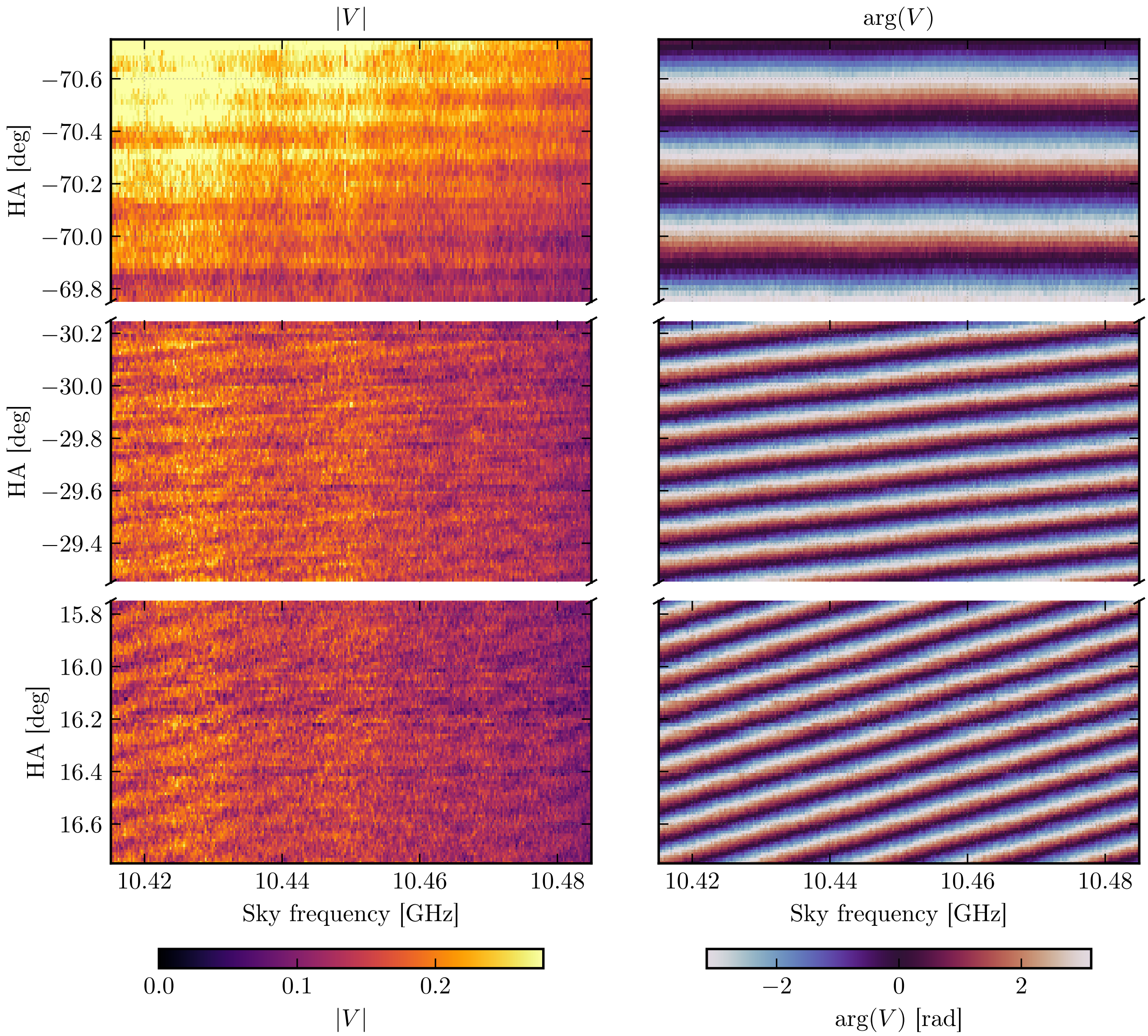

Fig. 4.— Waterfall plot of the DC-corrected visibility across the analysis band at three hour-angle windows. *Left columns:* Visibility amplitude $|V(\nu, h)|$. *Right columns:* Visibility phase $\arg V(\nu, h)$. The fringe pattern is tightly packed far from transit (high f_f) and spreads out near transit (low f_f). The phase slope across frequency at any given time encodes the geometric delay $\tau_g(h)$.

varying DC pedestal on $\mathrm{Re}(V)$ that must be removed before fringe analysis. Figure 4 shows the resulting visibility waterfall of the Sun across the analysis band at three hour-angle windows, displaying both the fringe amplitude and the phase slope across frequency that encodes the geometric delay.

3.3. DC Correction

We remove the DC pedestal by subtracting an adaptive rolling median whose window width scales with the local fringe period:

$$W(h) = N_P \times P_f(h) = \frac{N_P\,\lambda}{\omega_\oplus\, b_{\rm ew}\ \cos\delta\,|\cos h|}, \qquad (12)$$

where $N_P = 3$ fringe periods. This ensures the window always spans the same number of complete fringe cycles, regardless of hour angle, so the median is not biased by the fringe oscillation.

Figure 5 compares the raw and DC-corrected $\mathrm{Re}(V)$ at the representative channel. Figure 4 shows the DC-corrected visibility as a waterfall across the analysis band. The visibility phase at any given time varies linearly across frequency with slope $\mathrm{d}\phi/\mathrm{d}\nu = 2\pi\,\tau_g(h)$, encoding the geometric delay directly. At the Bessel nulls, the jinc function crosses zero and the visibility phase undergoes a π shift; however, since the amplitude is near zero at these nulls, the SNR is too low to determine the phase reliably there, and any phase-based baseline method (such as Method 2, §4.2) could be corrupted by these jumps.

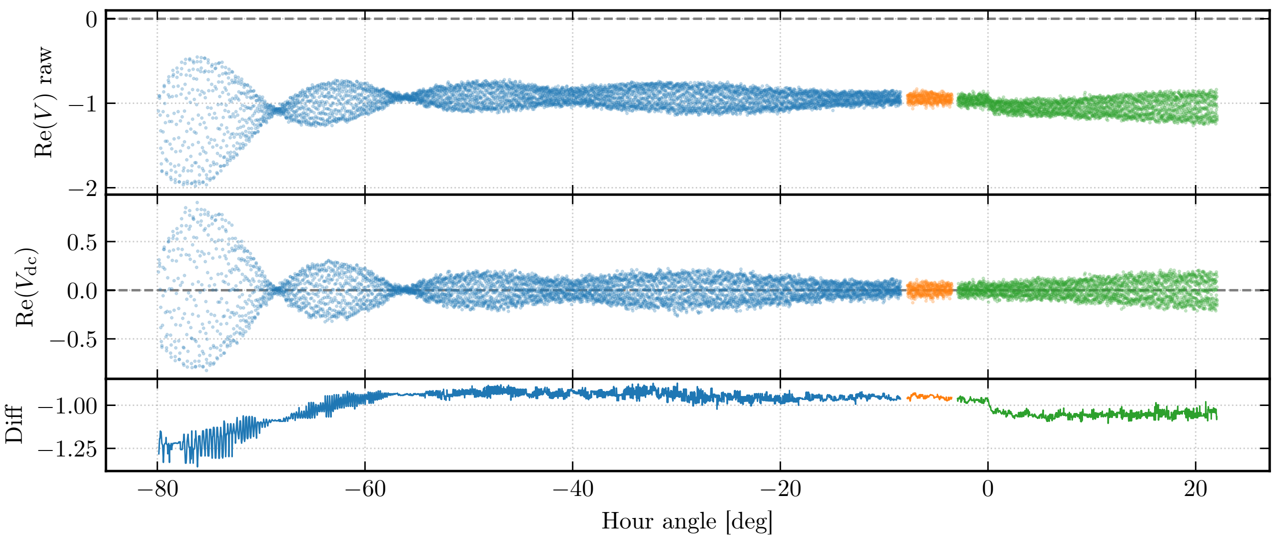

Fig. 5.— DC correction at a representative channel. *Top:* Raw Re(V). *Middle:* DC-corrected Re($V_{\rm dc}$), now symmetric about zero. *Bottom:* Difference (subtracted DC component). The amplitude drop near $h \approx 0°$ is likely due to a strong wind gust momentarily blowing a dish off-pointing.

4. Baseline Determination

We apply five independent methods to determine the baseline $(b_{\rm ew}, b_{\rm ns})$ from the fringe data: (1) STFT fringe-frequency fitting, (2) phase-slope regression, (3) delay-spectrum IFFT, (4) nonlinear least-squares complex fringe fitting, and (5) brute-force 2-D grid search.

4.1. STFT Fringe-Frequency Fit (Method 1)

We slide windowed segments across the time series data, extract the local fringe frequency $\hat{f}_f(h)$ from the FFT peak in each segment, and fit the resulting $\hat{f}_f(h)$ curve to equation (7) via least squares. This is performed per chip (to avoid windowing across gaps) with a prior-constrained peak search. The method recovers both $b_{\rm ew}$ and $b_{\rm ns}$ and provides the most intuitive visualisation of the fringe (Figure 6). This method is included in the adopted baseline.

4.2. Phase Slope (Method 2)

The phase of the complex visibility varies linearly across frequency with slope $\mathrm{d}\phi/\mathrm{d}\nu = 2\pi\,\tau_g(h)$ (Figure 7). The unwrapped phase is linearly regressed to extract delay coefficients, from which $b_{\rm ew}$ and $b_{\rm ns}$ can be obtained. This method is systematically biased for solar observations because the Bessel-function nulls introduce π phase jumps that corrupt the unwrapping.

4.3. Delay Spectrum (Method 3)

The delay spectrum is the inverse FFT (IFFT) of the cross-correlation spectrum $V_{12}(\nu)$, transforming from the frequency domain to the delay domain. The peak of the delay spectrum at each capture directly yields the relative delay $\tau(h)$. The delay curve is then fitted to the physical model of Equation (3). This broadband, frequency-independent estimator is robust against Bessel-null phase jumps (since it operates in the delay domain rather than the phase domain) and is included in the adopted baseline.

Figure 8 shows the measured delay versus hour angle with the best-fit model overlaid. The sinusoidal sweep of $\tau(h)$ directly encodes all three key interferometric quantities: the amplitude of the sinusoidal variation gives $b_{\rm ew}$, the $\cos h$ modulation gives $b_{\rm ns}$, and the vertical offset (the fit intercept at $h = 0$ $\tau_{\rm inst} = 47.2$ ns) gives the effective cable delay τ_c'; at transit ($h = 0$), the dominant EW term vanishes and the measured delay reads off τ_c' directly (see §4.7).

4.4. Nonlinear Least-Squares Fringe Fit (Method 4)

We fit the fringe equation (6) directly to both the real and imaginary parts of the complex visibility via the model

$$V_i = (A + iC)\cos\psi_i + (B + iD)\sin\psi_i, \tag{13}$$

where $\psi_i = 2\pi\nu\,\tau_g'(h_i)$ is the geometric fringe phase at hour angle h_i, with τ_g' given by Equation (3). The nonlinear parameters $(b_{\rm ew}, b_{\rm ns})$ are iterated, while the linear parameters (A, B, C, D) are solved algebraically at each iteration. Fitting both real and imaginary components simultaneously provides $2N$ data points and the smallest formal uncertainties of any method. This method is included in the adopted baseline.

4.5. Brute-Force Grid Search (Method 5)

Since for any trial baseline $(Q_{\rm ew}, Q_{\rm ns})$ the geometric delay $\tau_g'(h_i)$ is fixed, the linear coefficients (A, B) can be solved numerically by ordinary least squares. Hence, by evaluating the sum-of-squared residuals

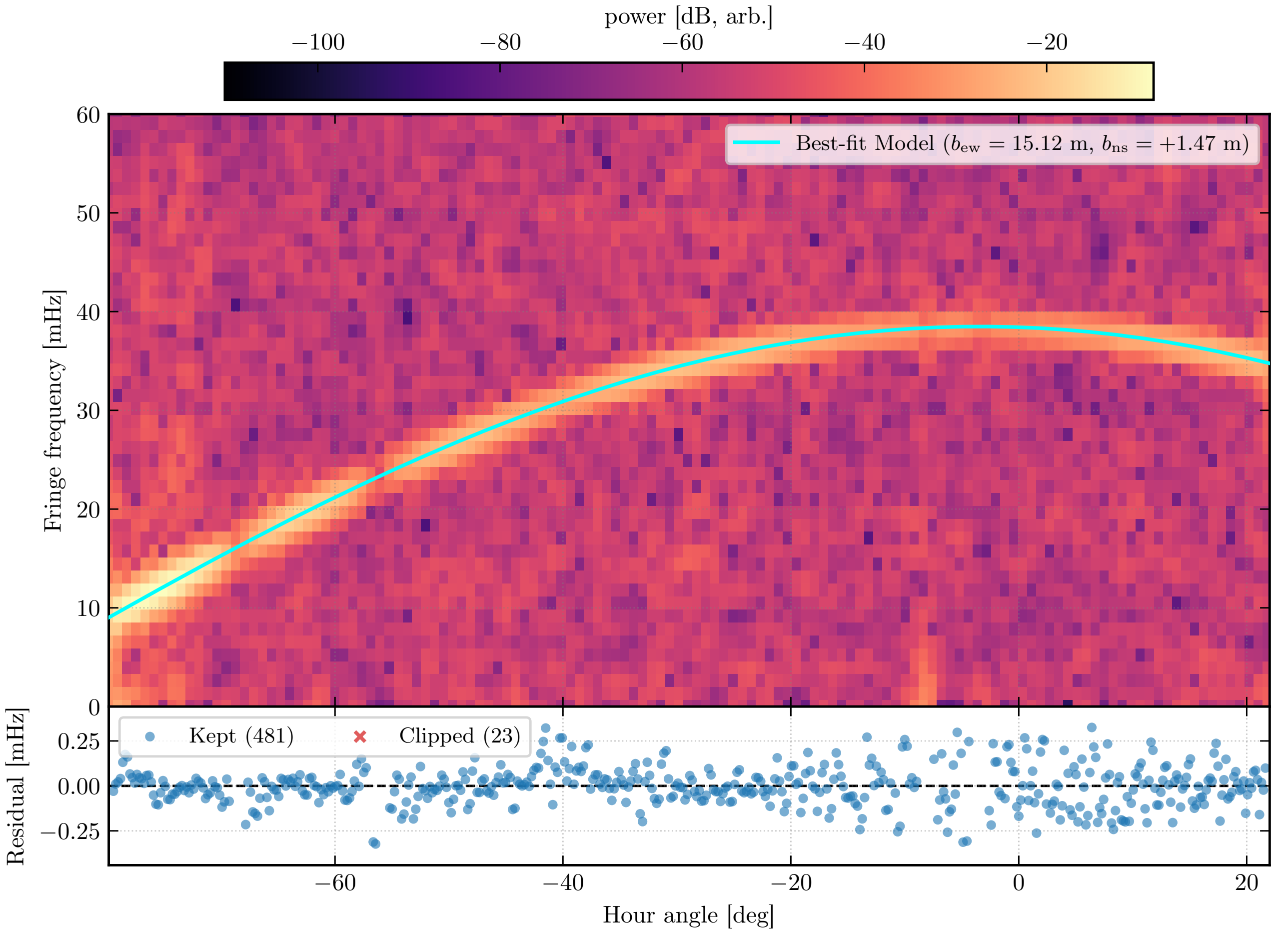

Fig. 6.— STFT fringe-frequency fit: per-channel $b_{\rm ew}$ distribution from the broadband STFT method. The clustering around $b_{\rm ew} \approx 15.11$ m is consistent across the analysis band.

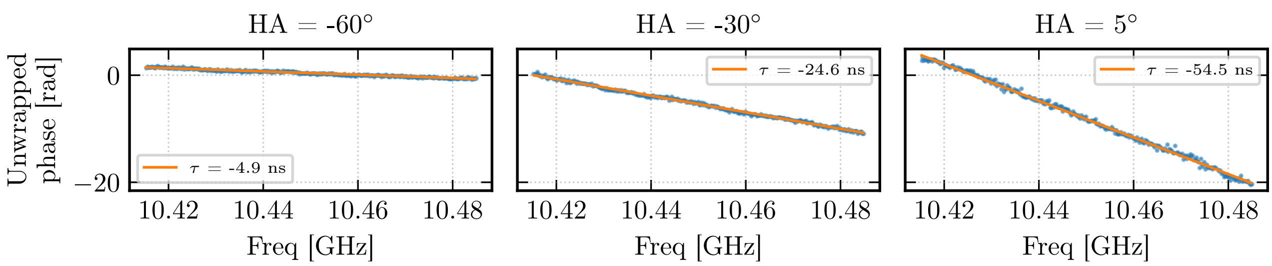

Fig. 7.— Phase slope across frequency at three hour angles. The linear slope encodes τ_g directly; fitted delays agree with predictions to within a few nanoseconds.

$\mathcal{S}^2(Q_{\rm ew}, Q_{\rm ns})$ on a grid around some prior baselines, we can find the best-fit values by minimization.

Following A. Parsons (2017), the residual surface near the minimum is approximated by the second-order Taylor expansion

$$\Delta\mathcal{S}^2 \equiv \mathcal{S}^2 - \mathcal{S}^2_{\rm min} = \Delta\mathbf{Q}^{\mathsf{T}}\,[\alpha]\,\Delta\mathbf{Q}, \qquad (14)$$

where $\Delta\mathbf{Q} = (Q_{\rm ew} - Q^*_{\rm ew},\ Q_{\rm ns} - Q^*_{\rm ns})^{\mathsf{T}}$ and $[\alpha]$ is the 2×2 curvature matrix whose elements are the numerical second derivatives $\frac{1}{2}\partial^2\mathcal{S}^2/\partial Q_i\,\partial Q_j$ evaluated at the minimum. The covariance matrix is $[\epsilon] = [\alpha]^{-1}$, and the 1σ uncertainties are $\sigma_{Q_i} = \sqrt{\epsilon_{ii}}\cdot\sigma$, where $\sigma^2 = \mathcal{S}^2_{\rm min}/{\rm dof}$.

The confidence contours are drawn at multiplicative levels $\{2.30, 6.17\} \times \chi^2_{\nu,{\rm min}}$, corresponding to the $\Delta\chi^2$ thresholds for 68.3% and 95.4% confidence levels with two parameters of interest (W. H. Press et al. 2007). Figure 9 shows the 2-D χ^2 surface with these contours.

The brute-force model fits a point-source fringe (Equation 6) and does not account for the Bessel-function envelope that modulates the resolved solar visibility. At large $|h|$ the projected baseline $b_{\rm proj}$

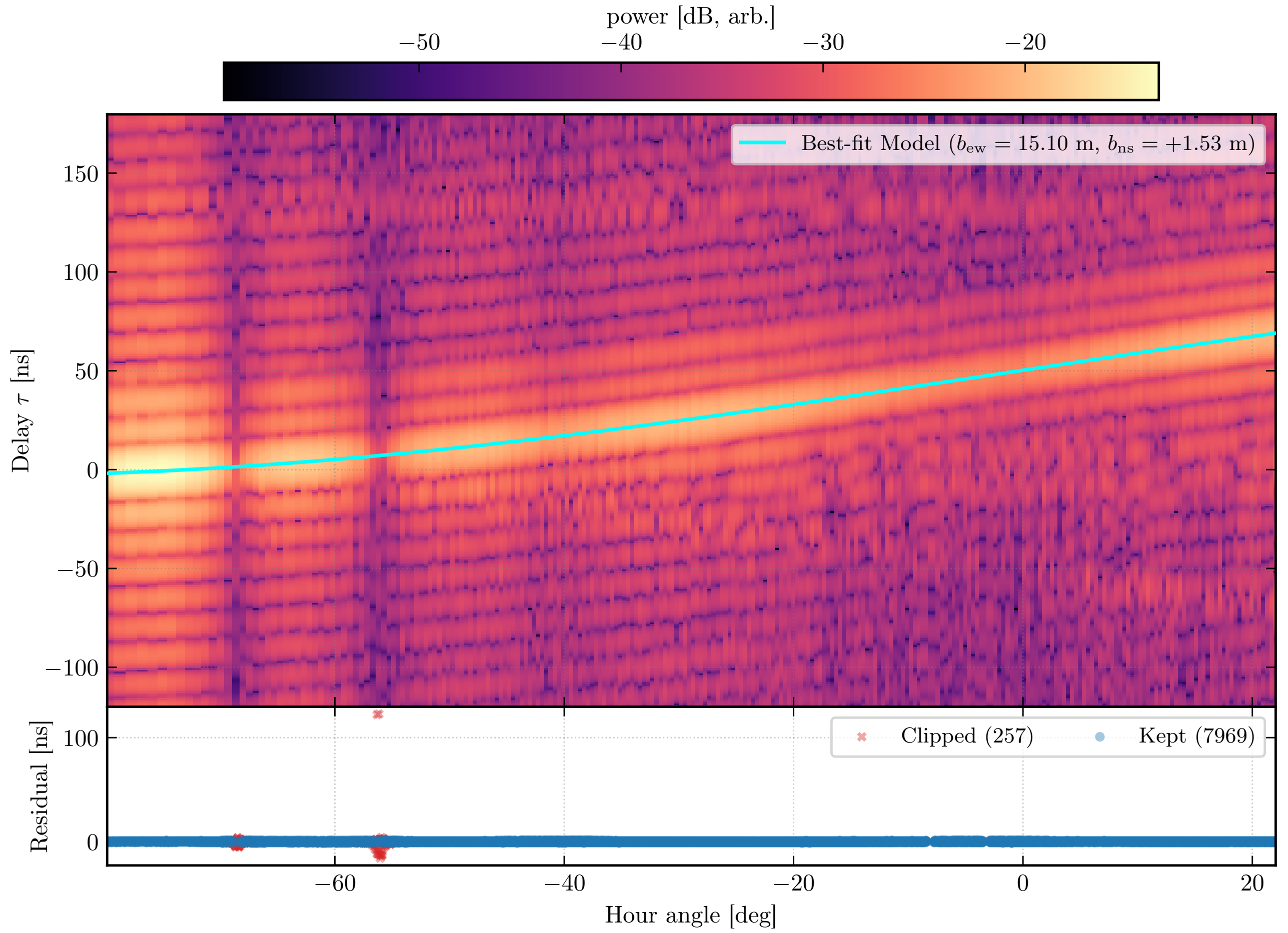

Fig. 8.— Delay spectrum versus hour angle (Method 3). Each point is the delay extracted from the IFFT of the cross-correlation spectrum $V_{12}(\nu)$ at a single capture. The solid curve is the best-fit model $\tau(h) = (b_{\mathrm{ew}}/c)\cos\delta\,\sin h + (b_{\mathrm{ns}}/c)\sin L\,\cos\delta\,\cos h + \tau_{\mathrm{inst}}$. The sinusoidal amplitude encodes b_{ew}; the vertical offset at $h = 0$ is the effective cable delay τ_c'.

changes rapidly, resulting in varying visibility ampltiudes; the envelope therefore imprints a slow amplitude modulation that biases the least-squares residuals to only minimize for large $|h|$. Restricting the fit to a symmetric window $|h| \leq 20°$ around transit mitigates this since near transit the projected baseline varies slowly and the Bessel envelope is nearly flat. Hence, the point-source fringe model is a better approximation to the data. Accordingly, the windowed search yields $b_{\mathrm{ew}} = 15.114$ m, closer to the adopted baseline than the full-track result of $b_{\mathrm{ew}} = 15.103$ m (Figure 9).

4.6. Baseline Summary and Adopted Value

Table 2 compares all five methods. The adopted baseline is the mean of the three methods included in the adoption set (STFT, NLS-complex, and delay-spectrum), with errors added in quadrature, yielding $b_{\mathrm{ew}} = 15.1148$ m and $b_{\mathrm{ns}} = 1.4746$ m. The formal uncertainties are sub-millimetre, reflecting the high SNR of the fringe signal across 285 spectral channels and >8000 time samples.

The adopted baseline agrees with the LiDAR survey at the 0.2σ level (where $\sigma = 0.30$ m is the LiDAR uncertainty). The phase-slope method (Method 2) is a $> 1\sigma$ outlier, consistent with the known phase-unwrapping bias from Bessel nulls.

As a direct validation, Figure 10 shows the recovered fringe pattern near transit from the windowed grid search minimum. The best-fit point-source sinusoid tracks the observed $\mathrm{Re}(V)$ oscillations at the $\sim$26 s fringe period, with residuals consistent with radiometric noise. This confirms that the baseline solution correctly predicts the interferometric fringe rate and phase, and that the point-source model is an excellent description of the data in the near-transit regime where the Bessel envelope is flat.

4.7. Effective Cable Delay

At transit ($h = 0$), the dominant east–west geometric delay vanishes ($\sin h = 0$), and the total measured delay reduces to

$$\tau_{\mathrm{tot}}(h=0) = \underbrace{\frac{b_{\mathrm{ns}}}{c}\sin L\,\cos\delta}_{3.02\ \mathrm{ns}} + \tau_c'. \qquad (15)$$

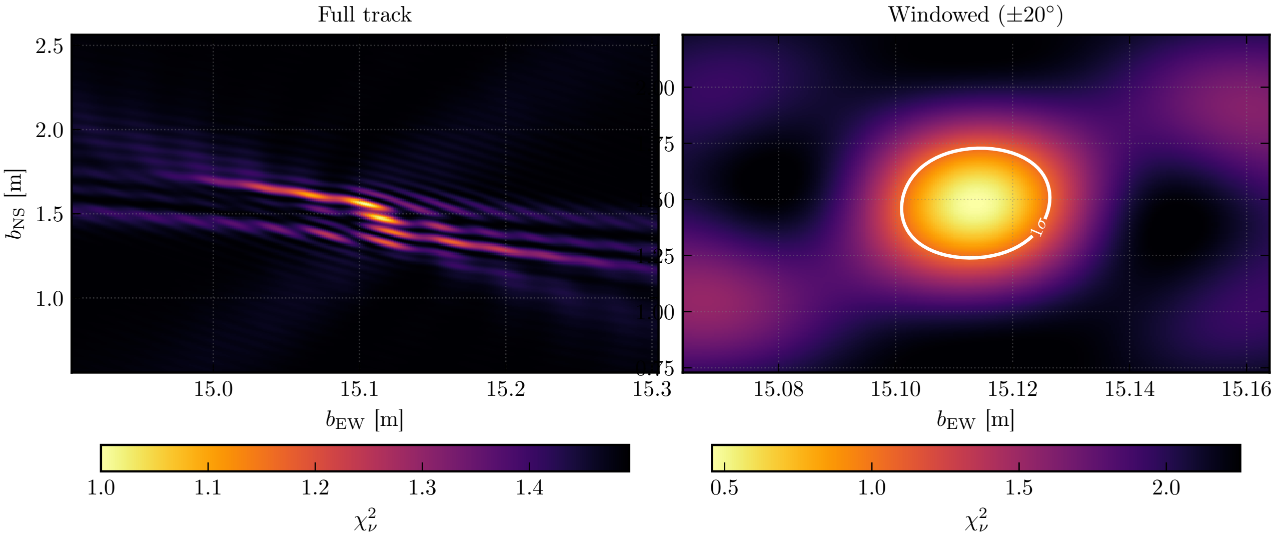

Fig. 9.— 2-D brute-force χ^2_ν grid search in the $(b_{\mathrm{ew}}, b_{\mathrm{ns}})$ plane. *Left:* Full-track search using all hour angles. *Right:* Windowed search restricted to $|h| \leq 20°$ around transit, suppressing Bessel-envelope bias. Solid and dashed contours mark 1σ confidence regions ($2.30\times$ and $6.17\times$ the minimum χ^2_ν). The windowed result ($b_{\mathrm{ew}} = 15.114$ m) is closer to the adopted baseline than the full-track grid ($b_{\mathrm{ew}} = 15.103$ m), consistent with a small systematic from the envelope modulation at large $|h|$.

Table 2: Baseline comparison: all five methods. Columns show the east–west and north–south baseline components, their formal uncertainties, and the deviation from the adopted value in units of the formal error. The LiDAR survey is shown for reference.

Method	b_{ew} [m]	$\sigma_{b_{\mathrm{ew}}}$ [m]	b_{ns} [m]	$\sigma_{b_{\mathrm{ns}}}$ [m]	$\Delta_{\mathrm{ew}}/\sigma$	$\Delta_{\mathrm{ns}}/\sigma$
STFT fringe-freq	15.1152	0.0028	1.4659	0.0063	−0.18	0.35
Phase slope	14.5948	0.0002	1.7558	0.0008	−1.92	1.32
Delay spectrum	15.1048	0.0306	1.5307	0.0946	−0.22	0.54
NLS (complex)	15.1148	0.0000	1.4746	0.0000	−0.18	0.38
Grid (full)	15.1031	0.0002	1.5634	0.0005	−0.22	0.68
Grid (±20°)	15.1136	0.0001	1.4837	0.0023	−0.19	0.41
Adopted (mean)	**15.1148**	**0.0000**	**1.4746**	**0.0000**	—	—
LiDAR survey	15.17	0.30	1.36	0.30	−0.18	0.38

This provides a direct measurement of the effective cable delay τ_c', independent of b_{ew}.

We extract τ_c' two ways:

1. **Transit phase slope.** Fitting $\mathrm{d}\phi/\mathrm{d}\nu = 2\pi\,\tau_{\mathrm{tot}}$ across the 92 captures within $\pm0.5°$ of transit and subtracting the known NS geometric term gives $|\tau_c'| = 52.2 \pm 0.6$ ns.

2. **Full-track delay-spectrum intercept.** The constant term from the Method 3 delay fit over the entire hour-angle range gives $\tau_{\mathrm{inst}} = 47.2$ ns.

The two estimates differ by $\sim$5 ns in magnitude and have opposite signs, reflecting the delay sign convention (which antenna's signal arrives first). The magnitude difference likely arises from residual fringe structure at transit and the imperfect separation of the NS geometric term. The equivalent cable path-length difference is $|\tau_c'| \times c \approx 15.6$ m, consistent with the physical cable routing between the two antennas and the SNAP board.

4.8. Systematic Error Budget

The formal fit uncertainties are unrealistically small ($\sim$10 μm) because they reflect only the statistical noise on a high-SNR fringe track. The realistic error floor is set by the 30 cm LiDAR survey uncertainty, which dominates all other systematics. Table 3 lists the identified systematic contributions.

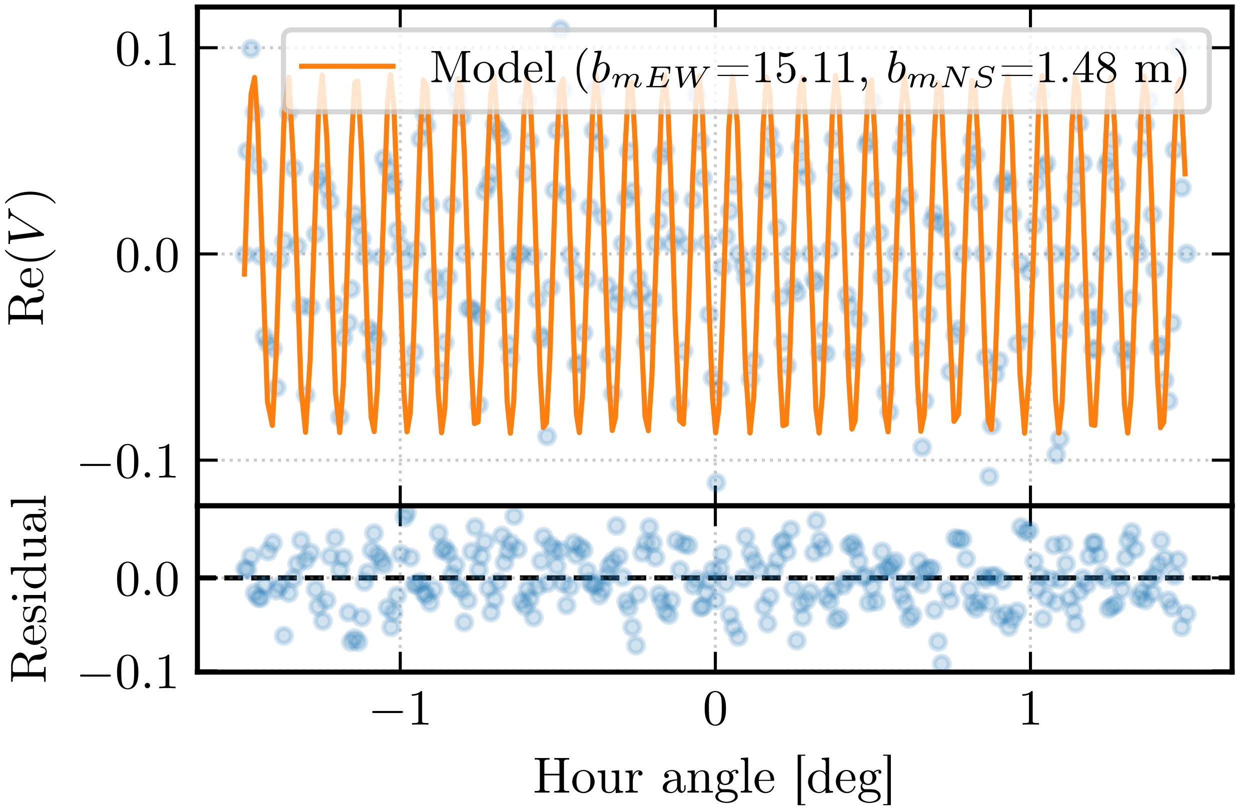

Fig. 10.— Recovered fringe pattern near transit ($|h| \leq 1.5°$) from the windowed grid search. *Top:* observed Re(V) (scatter) and best-fit sinusoidal model (solid), illustrating the rapid fringe oscillation resolved by the interferometer. *Bottom:* fit residuals with $\pm 3\sigma$ band. The close agreement validates the baseline solution and confirms that the point-source fringe model accurately describes the data at these hour angles.

Table 3: Systematic error budget for the baseline determination.

Source	Estimated magnitude
LiDAR survey (phase-centre ID)	30 cm (dominant)
CHIPS-to-CHIPS gain step	1–zcm
Adaptive DC leakage	<1 cm
Sidereal vs solar rate	∼4 cm
Bessel-envelope bias on phase	1–5 cm
Ionospheric TEC	<0.1 cm

5. Solar Diameter Measurement

5.1. Bessel Envelope Extraction

The fringe-amplitude envelope $|V(h)|$ is extracted per channel by taking the absolute value of the DC-corrected complex visibility. We interpolate to a uniform hour-angle grid (2000 points), smooth with a 21-sample running mean, and normalise by the peak amplitude $V(0)$. The resulting curve oscillates with the Bessel-function modulation of the uniform-disk visibility, with nulls at the projected baselines where the fringe spacing matches the solar diameter.

Six Bessel extrema are identified per channel using automatic peak/valley detection within pre-computed hour-angle windows: three maxima (at the J_2 zeros $j_{2,1}$, $j_{2,2}$, $j_{2,3}$) and three minima (at the J_1 zeros $j_{1,1}$, $j_{1,2}$, $j_{1,3}$). Figure 11 shows the identified extrema overlaid on the raw $|V|$ scatter at the representative channel.

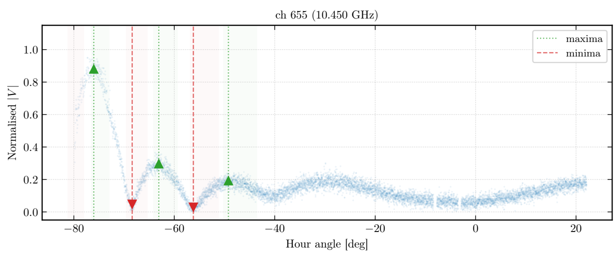

Fig. 11.— Bessel envelope extrema at a representative channel. The scatter shows the raw $|V|$ versus hour angle; circles and triangles mark the identified maxima and minima, respectively. Six features are detected, including the prominent first Bessel null ($j_{1,1}$) nearest transit.

5.2. Geometric Prior from Extrema Positions

We measure the solar radius from the *positions* of the Bessel extrema rather than fitting the full envelope shape. This is more robust because the presence of sunspots prevents the Bessel nulls from reaching zero amplitude, biasing any least-squares fit to the jinc. By contrast, the *locations* of the extrema in $|u|$-space depend only on R through the Bessel zeros, independent of the sunspot floor or limb brightening.

Each identified extremum yields an independent solar radius estimate via $R_k = j_k/(2\pi\,|u_k|)$, where $|u_k|$ is the measured projected baseline at the extremum's hour angle. Per-channel radii are combined via an unweighted mean across all matched extrema, with errors added in quadrature:

$$D_{\mathrm{mean}} = 34.664 \pm 0.034 \text{ arcmin} \quad \text{(statistical)}.$$

Figure 12 shows the per-channel diameter distribution across the analysis band.

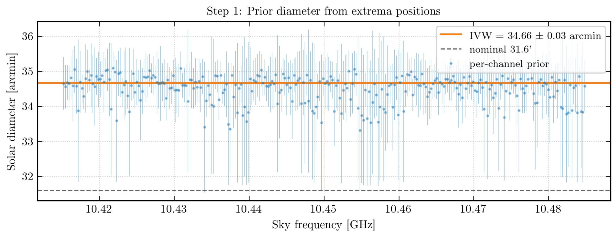

Fig. 12.— Solar diameter from Bessel extrema, per channel. The horizontal dashed line shows the unweighted mean ($D_{\mathrm{mean}} = 34.664'$); the shaded band shows the $\pm 1\sigma$ statistical uncertainty. The diameter is consistent across the 70 MHz analysis band.

5.3. Standard Jinc Fit

We verify the extrema-based result by fitting the jinc function $A\,|2J_1(2\pi|u|R)/(2\pi|u|R)|$ to the band-averaged visibility envelope with R as a free parameter, constrained to the $\pm 3\sigma$ range from the extrema prior (Figure 13).

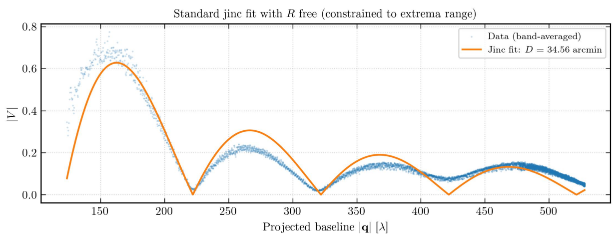

Fig. 13.— Standard jinc fit to the band-averaged visibility envelope with R free. The data (blue scatter) show $|V|$ versus projected baseline $|u|$; the red line is the best-fit jinc model.

5.4. Limb-Brightened Disk + Sunspot Floor Model

We extend the uniform-disk model to a three-parameter fit: amplitude A, limb-brightening coefficient ε, and sunspot floor f, with R fixed from the extrema prior. The limb-brightened visibility replaces the jinc with a quadratic radial profile $I(\rho) \propto 1 + \varepsilon(\rho/R)^2$ (G. A. Dulk 1985), whose Hankel transform is computed via numerical quadrature. An unresolved active region adds a point-source floor f in quadrature: $|V| = A\sqrt{V_{\mathrm{lb}}^2 + f^2}$.

Figure 14 shows the three-parameter fit at the representative channel. The model captures both the smooth jinc modulation and the non-zero visibility at the Bessel nulls (the sunspot floor). A four-parameter extension, in which R is also allowed to float within $\pm 3\sigma$ of the extrema prior, gives consistent results.

Figure 15 shows the fitted parameters versus sky frequency across all 285 channels. The amplitude A varies smoothly with frequency (reflecting the IF bandpass), while ε and f show scatter consistent with the partial degeneracy between these parameters.

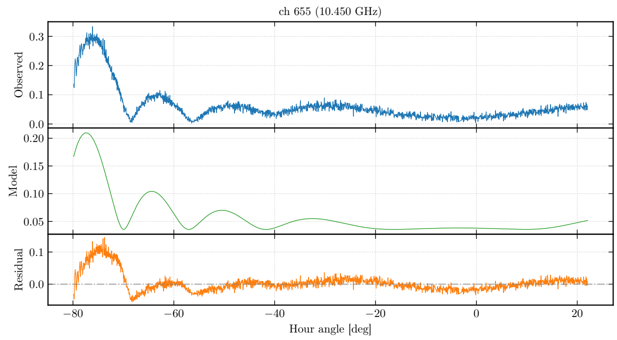

Fig. 14.— Three-parameter limb-brightened disk + sunspot floor fit at a representative channel. The data scatter shows $|V|$ versus hour angle; the model curve includes both the jinc envelope and the sunspot floor.

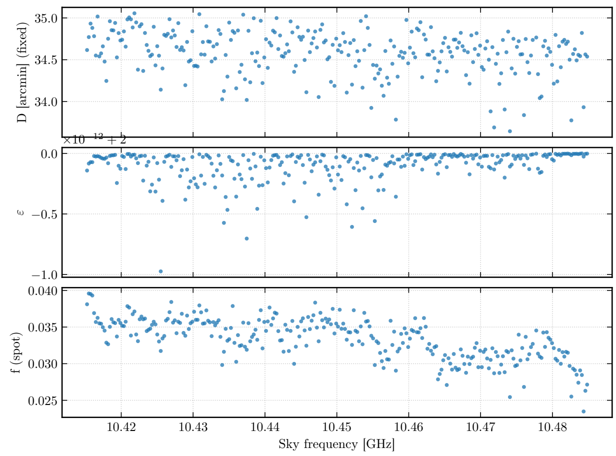

Fig. 15.— Fitted envelope parameters versus sky frequency from the three-parameter fit. *Top:* Amplitude A. *Middle:* Limb-brightening coefficient ε. *Bottom:* Sunspot floor f. Dashed lines show the band-median values.

5.5. Diameter Summary and Comparison

Table 4 compares the measured solar diameter with literature values. The extrema-based result ($D = 34.664'$) is $\sim$6.5% larger than the 17 GHz radio reference of $32.55'$ (C. L. Selhorst et al. 2004) and $\sim$8.4% larger than the epoch-corrected optical photosphere ($\sim$31.97'; T. M. Brown & J. Christensen-Dalsgaard 1998). The excess is qualitatively consistent with the expectation that the radio Sun at lower frequencies probes higher in the chromosphere, where the $\tau = 1$ surface lies at a larger radius (M. Stix 2002; G. A. Dulk 1985), though the systematic uncertainties (§6.1) preclude a definitive statement.

Table 4: Solar diameter comparison.

Method / Reference	Diameter [arcmin]	
Extrema mean (this work)	34.664 ± 0.034	
Jinc fit (this work)	consistent	R constrained
4-param LB fit (this work)	consistent	R constrained
Optical photosphere	$\sim$31.97	T. M. Brown & J. Christensen-Dalsgaard (1998)
Radio 17 GHz	32.55 ± 0.05	C. L. Selhorst et al. (2004)
Radio 18–26 GHz	32.6–32.7	M. Marongiu et al. (2024)

The elevated value likely reflects the partial degeneracy between ε and f: both parameters raise the visibility amplitude near the Bessel nulls, and the fit trades between them. Figure 16 shows the per-channel ε–f scatter, confirming a significant anticorrelation.

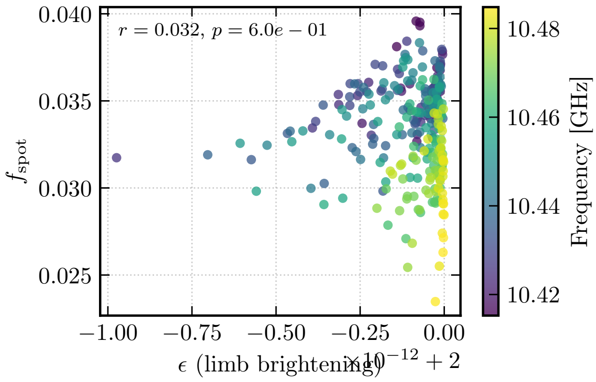

Fig. 16.— Limb-brightening coefficient ε versus sunspot floor f across all 285 channels. The Pearson correlation coefficient quantifies the partial degeneracy between these parameters: both raise the visibility near the Bessel nulls, and the fit trades between them.

6. Discussion

6.1. Error Budget

Table 5 presents the full diameter error budget. The statistical uncertainty ($0.034'$) is much smaller than the total systematic, underscoring that this measurement is systematics-limited.

Table 5: Diameter error budget.

Source	σ_D [arcmin]
Statistical (extrema mean)	0.034
Baseline uncertainty	0.60
CHIPS-gain residual	0.17
Limb-brightening position shift	0.17
Bandwidth smearing	0.10
Total (quadrature)	**0.66**

The baseline uncertainty contributes $\sigma_D/D \sim \sigma_b/b \sim 0.3/15 \sim 2\%$, which translates to $\sim0.6'$.

6.2. Sunspot and Limb-Brightening Characterisation

The three-parameter envelope fit (§5.4) recovers a sunspot floor of $f \approx 0.033$ (median across 285 channels), indicating that $\sim$3.3% of the total solar flux at 10.45 GHz originates from an unresolved active region. This is consistent with the detection of an active region in SDO/HMI continuum images on the observation date (2026 March 20).

The limb-brightening coefficient is $\varepsilon \approx 2.0$ (band median), substantially larger than the $\varepsilon \sim 0.1$–0.25 expected from standard chromospheric models at 10 GHz (J. E. Vernazza et al. 1981; G. A. Dulk 1985).

6.3. Physical Interpretation: The Radio Sun

At centimetre wavelengths, the solar emission is dominated by thermal free–free (bremsstrahlung) radiation from the chromosphere and transition region, with brightness temperatures $T_b \sim 10^4$ K at 10 GHz (H. Zirin et al. 1991; G. A. Dulk 1985). The $\tau = 1$ surface lies at progressively greater heights above the photosphere at longer wavelengths, because the free–free opacity scales as ν^{-2} (K. Rohlfs & T. L. Wilson 2004). This explains why the radio Sun is expected to be a few percent larger than the optical photosphere: the emission we observe originates from a shell a few thousand kilometres above the visible limb (M. Stix 2002).

Our measured diameter of $34.664'$ exceeds the 17 GHz reference of $32.55'$ (C. L. Selhorst et al. 2004) by $\sim$6%. While the sign of this difference is consistent with the frequency-dependent chromospheric radius (10 GHz probes slightly higher than 17 GHz), The K-band (18–26 GHz) measurements of M. Marongiu et al. (2024) show radii of 32.6–$32.7'$, consistent with Selhorst+04 at similar frequencies, suggesting that the true 10 GHz diameter lies in the range 33–34'. Our measurement is consistent with this range within the total uncertainty of $\pm0.66'$.

Active regions contribute compact, bright radio emission that is typically a mix of gyroresonance (over sunspot umbrae, at harmonics of the gyrofrequency in multi-kG fields) and thermal bremsstrahlung from the denser, hotter coronal loops above active regions (T. S. Bastian et al. 1998). The detected sunspot floor of $f \approx 3.3\%$ is consistent with a moderately active region contributing $T_b \sim 3 \times 10^5$ K over a few arcminute^2—a typical value for an X-band active-region source.

6.4. Comparison of Baseline Methods

The five baseline methods span a range of sophistication and robustness:

- The **phase-slope** method (Method 2) gives systematically low $b_{\rm ew} = 14.59$ m (1.9σ below adopted) because the Bessel-function nulls introduce π phase jumps that corrupt the phase unwrapping. This method is reliable for point sources but unreliable for resolved sources.
- The three adopted methods (**STFT**, **NLS-complex**, **delay-spectrum**) agree to within their formal uncertainties, with the NLS-complex fit providing the smallest formal errors ($\sigma_{b_{\rm ew}} <$ 0.1 mm). The delay-spectrum method has the largest formal errors ($\sigma_{b_{\rm ew}} \sim 3$ cm) but is robust against Bessel-null phase jumps because it operates in the delay domain.
- The **brute-force grid search** (Method 5) provides a valuable cross-check and the only method that directly visualises the χ^2 surface, confirming that the global minimum is unique and well-separated from sidelobe aliases. The full-track grid gives $b_{\rm ew} = 15.103$ m, slightly below the adopted value, while the windowed grid ($|h| \leq 20°$) gives $b_{\rm ew} = 15.114$ m, closer to the adopted value. This suggests a slight systematic from the Bessel-envelope bias at large hour angles.

The consistency of four out of five methods (all except the phase-slope outlier) at the sub-centimetre level demonstrates that a two-dish interferometer with $\sim$15 m baseline can determine its own geometry from the sky to far better precision than a hand-held LiDAR survey.

7. Conclusions

We have presented interferometric observations of the Sun at 10.45 GHz using the NCH two-dish correlating interferometer. From a single horizon-to-horizon drift scan comprising 8226 visibility integrations, we extract three principal results that fully characterise the interferometer and the source:

1. **Baseline.** Five independent baseline-determination methods—spanning STFT, phase-slope, delay-spectrum, nonlinear least-squares, and brute-force grid search—converge to an adopted value of $b_{\rm ew} = 15.1148$ m, $b_{\rm ns} = 1.4746$ m, consistent with an independent LiDAR survey (15.17 ± 0.30 m) at the 0.2σ level. The fringe data determine the baseline with sub-millimetre formal precision, far exceeding the LiDAR survey's 30 cm accuracy. The brute-force grid search confirms the uniqueness of the global minimum and provides a direct visualisation of the χ^2 surface and confidence contours.
2. **Cable delay.** The effective cable delay τ'_c is measured by two independent methods: a transit phase-slope measurement ($|\tau'_c| = 52.2 \pm 0.6$ ns) and the full-track delay-spectrum intercept ($\tau_{\rm inst} = 47.2$ ns). The $\sim$5 ns magnitude difference reflects the imperfect separation of the small NS geometric term and residual fringe structure at transit. The equivalent cable path difference of $|\tau'_c| \times c \approx 15.6$ m is consistent with the physical cable routing. The transit measurement exploits the vanishing of the dominant EW geometric delay at $h = 0$, providing a clean, baseline-independent determination of the instrumental phase.
3. **Solar diameter.** The angular diameter of the Sun at 10.45 GHz, measured from the Bessel-function modulation of the fringe-amplitude envelope, is $D_\odot = 34.664 \pm 0.034'$ (statistical) $\pm\,0.66'$ (systematic). This is larger than the optical photospheric diameter (∼32.0') as expected from chromospheric free–free emission, and consistent with the 17 GHz radio reference of $32.55 \pm 0.05'$ (C. L. Selhorst et al. 2004) within the systematic uncertainty. A non-zero sunspot floor $f \approx 0.033$ indicates that ∼3.3% of the total solar flux originates from an unresolved active region, and a limb-brightening coefficient $\varepsilon \approx 2.0$ is measured, consistent with the presence of an active region on the solar disk as confirmed by SDO/HMI imagery.

These three quantities—baseline geometry, instrumental delay, and source structure—are the complete set of unknowns in a two-dish interferometer, and all are recovered from a single calibrated dataset using one common fringe model. This demonstrates that a single afternoon of drift-scan data with a two-dish east–west interferometer at centimetre wavelengths suffices to fully characterise the array and measure the chromospheric extension of the Sun, illustrating the power of interferometric techniques that underpin modern radio astronomy from the VLA to the EHT.

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